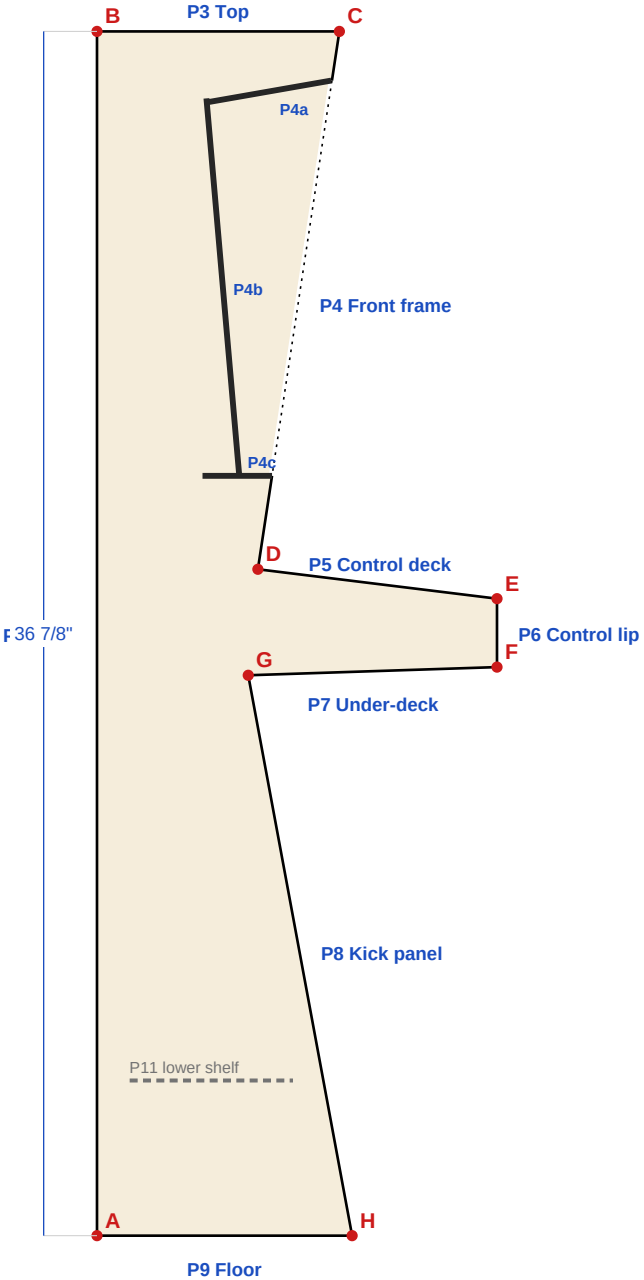
SUPERCADE - cardboard edition

Half scale. Printable plan set, built from the original Supercade CAD files.



P1 SIDE PANEL (x2) - side view, back on the left

AT A GLANCE

Overall height	36 7/8"
Overall depth	12 1/4"
Inside width	14 1/2"
Outside width	14 1/2" + 2 x side thickness
Control deck height	~20" (7 deg tilt)
Screen	13.3" portable USB-C monitor
Material	cardboard body, 5 mm Foamex deck plate
Fasteners	kids' plastic cardboard screws + glue

WHAT'S IN THIS PACKET

1 This overview
2 Side panel - measure-and-mark drawing
3 Cut list (every part, every size)
4 Cross panels - fold-flange layout
5 Control panel layout + hole schedule
6 Shopping list
7 Assembly steps

1:1 print-and-tape templates in half-scale/templates/:
control-panel-1to1.pdf
speaker-cutouts-1to1.pdf
side-panel-1to1.pdf

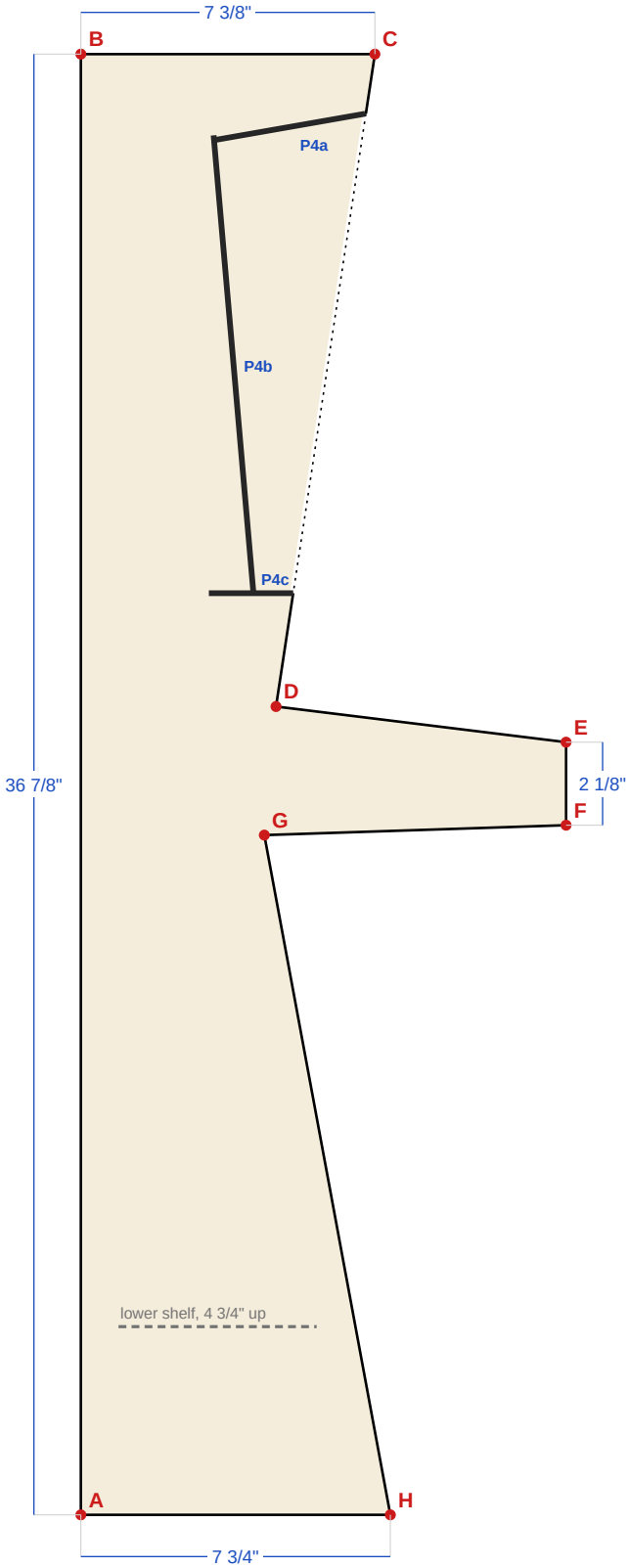
HOW THE CARDBOARD VERSION WORKS

The original is 1/2" plywood held together with wood screws driven into panel edges. Cardboard edges can't hold a screw, so every cross panel (P2-P9) gets a 1 1/2" flange folded down on each end. The flange lies flat against the inside of the side panel and a plastic screw + nut goes through both layers, face to face. That's exactly the joint kids' cardboard screw kits are made for.

Everything is the original at 50% - except the controls. Arcade buttons and joysticks don't shrink, so the deck has its own compact 2-player layout (page 5).

P1 Side panel (make 2)

Draw it straight onto the cardboard with a tape measure and a long straightedge. Every edge is a straight line.



MARK THESE POINTS

Lay the sheet flat. Draw the BACK edge first as a long straight line (A to B). Every point is measured UP from the floor end along that line, then OUT square to it (use a set square or a big book).

Pt	Up from floor	Out from back	Is
A	0" (0.00)	0" (0.00)	floor, back corner
B	36 7/8" (36.87)	0" (0.00)	top, back corner
C	36 7/8" (36.87)	7 3/8" (7.42)	top, front corner
D	20 3/8" (20.40)	4 7/8" (4.92)	screen meets deck
E	19 1/2" (19.50)	12 1/4" (12.24)	deck, front edge
F	17 3/8" (17.41)	12 1/4" (12.24)	lip, bottom
G	17 1/8" (17.16)	4 5/8" (4.63)	under deck, inner
H	0" (0.00)	7 3/4" (7.81)	floor, front corner

INSIDE LINES (pencil only - where the recess parts go)

S1	34 3/4" (34.70)	3 3/8" (3.37)	P4a speaker shelf, back
S2	35 3/8" (35.36)	7 1/8" (7.17)	P4a speaker shelf, front
M1	23 1/4" (23.26)	4 3/8" (4.36)	P4b screen board, bottom
M2	34 7/8" (34.81)	3 3/8" (3.35)	P4b screen board, top
R1	23 1/4" (23.26)	3 1/4" (3.23)	P4c recess floor, back
R2	23 1/4" (23.26)	5 3/8" (5.36)	P4c recess floor, front

Join the dots in order A-B-C-D-E-F-G-H-A with straight lines. Draw the inside lines S1-S2, M1-M2 and R1-R2 on the INSIDE face: the recess parts' flanges line up on them.

Check your work: D-E should be 7 3/8" long and C-D 16 5/8". If not, a point is off.

Second side: cut the first one out, flip it over, and trace it. The two sides are mirror images, so tracing the flipped panel gives you the correct other hand automatically.

Flutes UP AND DOWN (parallel to A-B): that direction carries the screen.

Stiffness: glue a second layer to each side (flutes crossed), OR glue 1 1/2" wide ribs of cardboard on the inside, keeping the 1 1/2" strip along every edge free for the flanges. See the screw-length note on page 6 before you pick.

Each side is 36 7/8" x 12 1/4". Trace the second from the first, flipped over.

Cut list

Cross panels are 14 1/2" between the sides + a 1 1/2" fold flange each end = 17 1/2" wide blanks.

Part	Name	Qty	Cut size (L x W)	Edge	Notes
P1	Side panel	2	36 7/8" x 12 1/4"	-	Shape on page 2. Mirror pair.
P2	Back	1	36 7/8" x 17 1/2"	A-B	Cut a 6" x 10" service door low down (optional).
P3	Top	1	7 3/8" x 17 1/2"	B-C	
P4	Front frame	1	16 5/8" x 17 1/2"	C-D	ONE continuous flat piece. Cut an opening 13" wide, from 1 1/2" to 13 3/4" down from the top (C) edge. The recess sits behind it.
P4a	Speaker shelf	1	3 7/8" x 17 1/2"	S1-S2	Recess ceiling. Two 2 5/8" x 2" speaker holes, pointing DOWN at the players.
P4b	Screen board	1	11 5/8" x 17 1/2"	M1-M2	Centred window 11 5/8" x 6 1/2" (13.3" monitor visible area; measure yours). Leans back 5 degrees; the monitor bolts or straps on behind it.
P4c	Recess floor	1	2 1/8" x 17 1/2"	R1-R2	Bottom of the recess, level with the top of the frame's lower lip.
P5	Control deck	1	7 3/8" x 17 1/2"	D-E	1 layer of cardboard + the Foamex plate (P12) on top. Under each joystick, cut an opening in the cardboard just bigger than the stick's mounting plate, so the stick bolts to the Foamex only.
P6	Control lip	1	2 1/8" x 17 1/2"	E-F	Front face under the players' wrists.
P7	Under-deck	1	7 5/8" x 17 1/2"	F-G	
P8	Kick panel	1	17 1/2" x 17 1/2"	G-H	Leg room slope. Optional access door.
P9	Floor	1	7 3/4" x 17 1/2"	H-A	Double layer. This carries the ballast.
P10	Screen shelf	1	13 3/4" x 17 1/2"	-	Folds into a 5" deep x 1 1/2" thick box beam (page 4). Sits under the monitor; set its height on the dry-run.
P11	Lower shelf	1	13 3/4" x 17 1/2"	-	Same box beam, at 4 3/4" up. Holds the computer and the ballast.
P12	Deck top plate	1	7 3/8" x 14 1/2"	-	5 mm Foamex (PVC foam board). Tape the 1:1 template to it, cut every hole, then drill through into P5.
P13	Corner strips	8	1 1/2" x 14 1/2"	-	Folded into L-angles and glued into inside corners that feel loose.

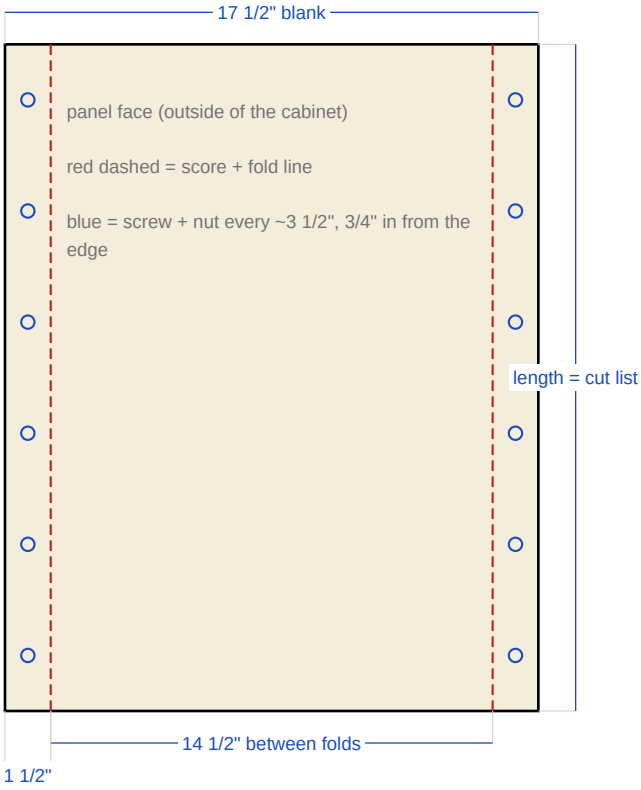
Cardboard: 4-5 big boxes. The longest parts (sides + back) are about 37" long, so you need at least one box with a panel that long - TV, bike and wardrobe boxes are.

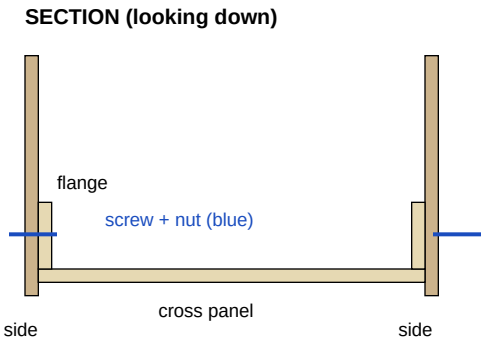
Every size is measured on the CAD edge. Cut cross panels 1/4" long and trim during the dry fit: cardboard is thicker than the pencil line, so corners will need a nibble.

Flutes: on the cross panels, run them ACROSS the 17 1/2" width so the flange fold goes along the flutes (cleaner fold, stiffer span).

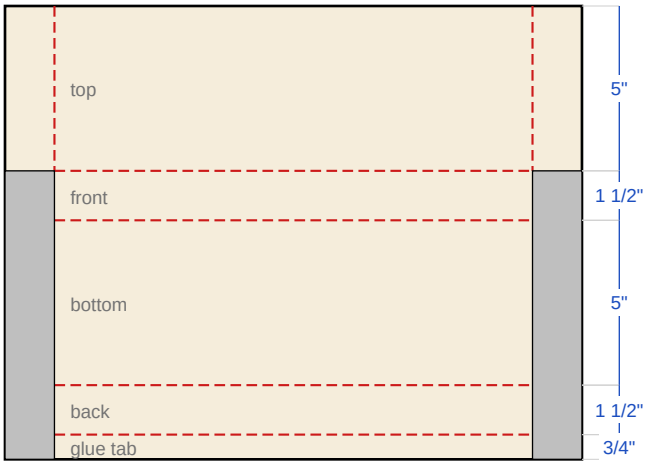
Cross panels - fold-flange layout

The same pattern for P2-P9. Score, fold 90 degrees, screw to the side panels.





P10 / P11 BOX-BEAM SHELF

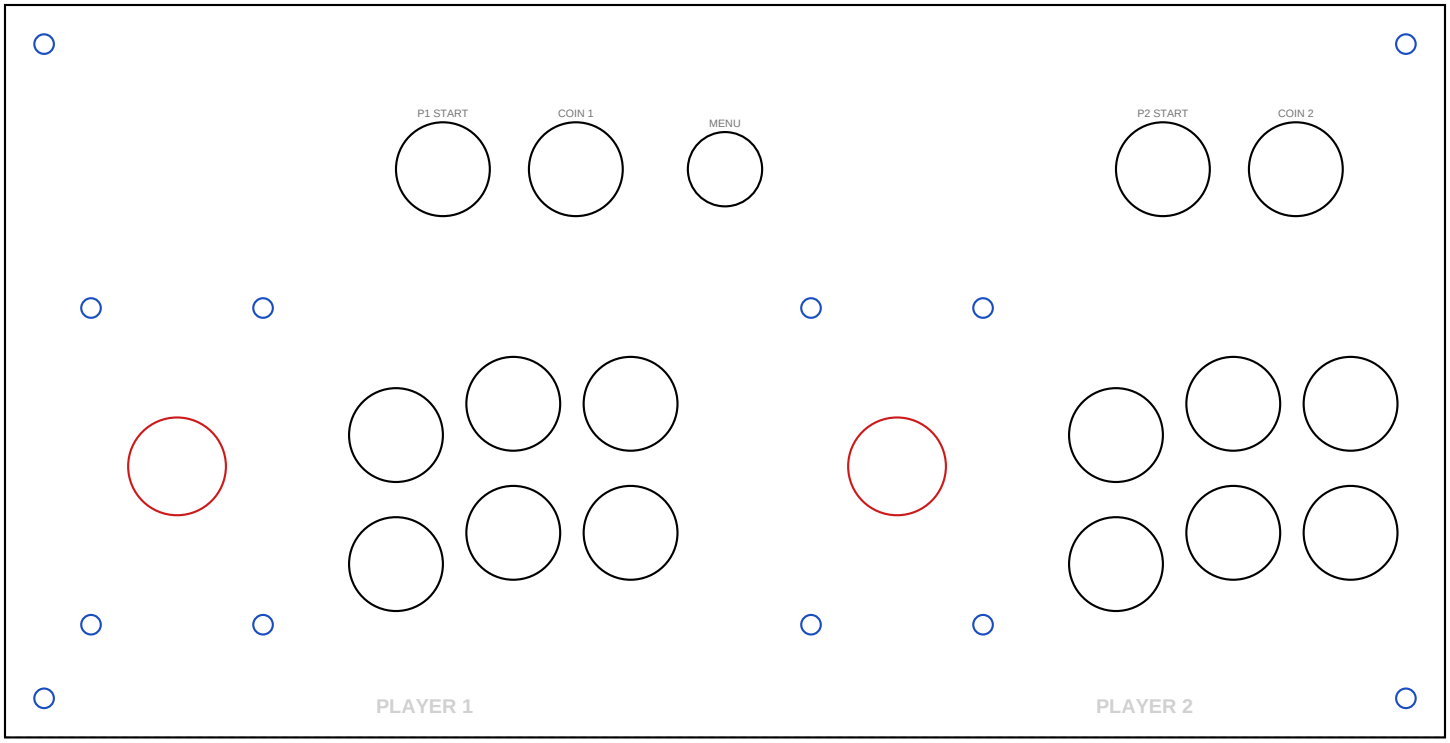


A 13 3/4" x 17 1/2" blank. Cut away the two grey strips (1 1/2" x 8 3/4") so the end flanges only remain on the 5" top. Score every red line. Fold the rest under into a closed 5" x 1 1/2" tube and glue the tab to the underside of the top. A closed tube is what makes a cardboard shelf stiff enough to hold a monitor. The end flanges screw to the side panels.

Screen shelf: set its height so the monitor sits centred behind the P4b window. Stand the monitor on it with its own stand removed, and strap the monitor back to the shelf or to P2 so it can't slide forward.

P5 Control panel layout

compact 2-player layout for the scaled deck. Print templates/control-panel-1to1.pdf for the real-size version.



dashed = full deck P5 (14 1/2" x 7 3/8"). Template is the whole deck: line its edges up with P5. Players stand at the bottom edge.

HOLE SCHEDULE

2	**25 mm**	Joystick shafts. The stick's plate bolts on underneath the deck.
12	**24 mm**	Action buttons. Use SCREW-IN buttons with a plastic nut - snap-in buttons only grip thin metal/acrylic.
2	**24 mm**	Coin buttons (Select on most emulators).
2	**24 mm**	Player Start buttons.
1	**19 mm**	Aux: menu / exit / volume.
12	**5 mm**	Bolt holes: joystick plates (4 per stick - CHECK against your own stick, spacing varies) and deck corners.

CUTTING ROUND HOLES

Tape the 1:1 template down. Push an awl (or the screw kit's punch tool) through every crosshair. Remove the paper, draw each circle with a compass, then cut with a sharp craft knife in several light passes. Faster and neater: a hole saw or step drill bit of the right size, run by a grown-up.

Cut small first and open up: the button should need a firm push, and the plastic nut does the rest. Keep the deck thin where the controls go (about 10 mm max): screw-in buttons and joysticks are made for thin panels.

Shopping list

Search terms, not specific listings - check sizes and reviews before you buy.

CARDBOARD + FASTENING

☐ **Big cardboard boxes**	4-5 large ones (TV, bike or wardrobe boxes). The longest parts are 37" long.
☐ **Kids' cardboard construction kit with screws + nuts**	e.g. a 400-piece kit with safe saw + crease roller. ~80 screws get used, so one kit is plenty.
☐ **5 mm PVC foam board ("Foamex"), A3 / 12 x 18" is enough**	For the control deck top plate. Cuts with a craft knife, holds buttons firmly.
☐ **PVA / white glue + hot glue gun**	Glue the laminations and the box-beam shelves.
☐ **Nylon or steel M4/M5 x 20 mm bolts + nuts + washers**	8 for the joysticks, 4 for the deck plate corners.
☐ **Wide packing tape / gaffer tape**	Edge-protecting the corners and the bottom edge.

CONTROLS + ELECTRONICS

☐ **"Zero delay" 2-player arcade kit with 24 mm buttons**	Needs 2 sticks + 16 x 24 mm buttons (12 action + 2 start + 2 coin). If the kit only has 30 mm, buy a 24 mm pack. SCREW-IN type.
☐ **13.3" portable USB-C monitor**	~0.7 kg. Built-in speakers are common, so check and skip separate ones.
☐ **Computer**	Raspberry Pi 4/5 with an arcade front-end, or an old laptop behind the cabinet.
☐ **Small USB speakers (optional)**	Only if the monitor has none. Two 2 5/8" x 2" oval cut-outs are in the templates.
☐ **Weight for the base**	A 2 kg bag of rice or a couple of books on the lower shelf.

TOOLS

☐ **Metal ruler (1 m / 40") + set square**	For drawing the side panels.
☐ **Craft knife + cutting mat**	Adults do the long cuts and the Foamex. Kids use the kit's safe saw.
☐ **Crease roller (in the kit) or a blunt butter knife**	For scoring fold lines.
☐ **Compass, pencil, tape measure**	Or a drill with 24 mm + 25 mm hole saws / a step bit for neat holes.

SCREW LENGTH - READ BEFORE YOU LAMINATE

Kids' plastic cardboard screws are short: they're made to join two layers of ordinary box cardboard. Before cutting anything, screw a scrap of your flange cardboard to a scrap of your side-panel cardboard, nut on. If it won't reach through a laminated (2-layer) side, keep the sides single-layer and stiffen them with glued ribs instead (page 2), or use nylon nuts and bolts at those joints.

Assembly steps

Dry-fit with screws only, glue once everything lines up.

1. **Test the screws**
   Screw scrap to scrap in the exact layers you'll use (page 6 note).

2. **Draw + cut the sides (P1)**
   Page 2. Cut one, flip it, trace the second. Laminate or rib them. Label the insides 'LEFT' and 'RIGHT'.

3. **Cut + score cross panels (P2-P9)**
   Page 3 sizes, page 4 flanges. Score with the flutes, fold every flange to 90 deg.

4. **Build the control deck (P5 + P12)**
   Cut every hole in the Foamex plate from the 1:1 template, lay it on P5 and mark through. Cut P5's holes plus a bigger opening under each stick. Bolt plate + P5 together at the corners, then mount sticks and buttons on the bench.

5. **Stand the back up (P2)**
   Lay one side flat, inside face up. Place P2 along edge A-B, flanges down against the side. Punch + screw.

6. **Floor + top (P9, P3)**
   Add P9 on H-A and P3 on B-C. You now have a rigid 'C'. Add the second side on top and screw all three.

7. **Stand it up + add the front**
   Stand the cabinet up. Screw in P8, P7, P6 and P5 (control deck, pre-wired). Then the recess, painted black first: P4c floor, P4b screen board, P4a speaker shelf. Last, P4 - the front frame - as one flat piece over the lot.

8. **Shelves (P10, P11)**
   Fold the box beams, screw their flanges in: lower shelf at the marked line, screen shelf so the monitor centres in the window.

9. **Weight + anchor**
   Put a 2 kg bag of rice on the lower shelf BEFORE the monitor goes in.

10. **Electronics**
    Computer on the lower shelf, encoder under the deck, monitor on its shelf. Run cables out through the service door.

11. **Glue + finish**
    Once it all works, back out one screw at a time, run glue into the joint, re-screw. Tape the edges, then paint or wrap it. Put Supercade art on the sides!

SAFETY. Small, but still top-heavy with the monitor in: weight the lower shelf, and stand it against a wall. Knife and Foamex cutting is for adults; kids use the kit's safe saw. Don't block the monitor's or computer's vents.